CLASS X MATHEMATICS • CHAPTER 13

Statistics

"Exhaustive point-by-point presentation covering grouped data mean (Direct, Assumed Mean, Step-Deviation), mode, median, cumulative distributions, and the empirical relationship directly from NCERT jemh113.pdf."

Akash Srivastva (www.akashmaths.online)

Measures of Central Tendency: From Ungrouped to Grouped Data

Recall from Class IX: Ungrouped Foundations

In Class IX, data was analyzed in ungrouped form or simple frequency tables. You represented distributions through **bar graphs**, **histograms**, and **frequency polygons**, and studied the three primary numerical representatives:

- ✓ **Mean ($\bar{x}$):** The numerical arithmetic average of all observed values.
- ✓ **Median:** The physical middle-most observation dividing sorted data in half.
- ✓ **Mode:** The observation with the single highest occurring frequency.

Why Grouping is Essential in Real Life

Real-world datasets (census data, national exam results, industrial lifetimes) contain thousands of data points. Condensing data into **continuous class intervals** is vital to uncover macro trends.

UNGROUPED MEAN FORMULA (CLASS IX RECALL)

$$\bar{x} = (\Sigma f_i x_i) / (\Sigma f_i)$$

Where observation x_i occurs with frequency f_i and Σ represents summation

The Class X Paradigm Shift

When raw observations are grouped into intervals like [10–25), individual observation identities are lost! We must now introduce:

- ✓ **Class Mark (x_i):** Central midpoint representing the entire class.
- ✓ **Modal Class:** Interval containing the peak frequency.
- ✓ **Cumulative Frequency (cf):** Running cumulative counts to locate the median.

⚡ Quick Box • Central Tendency Fundamentals

Q1 What are the three numerical measures of central tendency studied in statistics, and what single distinct property does each measure represent?

CONCEPT RECALL

Q2 Why can't we directly apply the simple ungrouped formula $\bar{x} = (\Sigma x_i) / n$ to grouped data where observations are given inside class intervals?

CONCEPTUAL SHIFT

Q3 If an observation value falls exactly on the boundary between two intervals (e.g., mark 40 for classes 25–40 and 40–55), into which class is it conventionally placed?

CLASS CONVENTION

Pages 173–174 (Textbook)

Mean of Grouped Data: The Direct Method

The Midpoint (Class Mark) Assumption

Because individual values inside an interval are unknown, we assume the frequency of each class is centered around its **midpoint (class mark x_i)**:

CLASS MARK DEFINITION

Class Mark (x_i) = (Upper Limit + Lower Limit) / 2

Example: For class 10–25, $x_i = (10 + 25) / 2 = \mathbf{17.5}$; for 25–40, $x_i = \mathbf{32.5}$.

DIRECT METHOD MEAN FORMULA

$$\bar{x} = (\Sigma f_i x_i) / (\Sigma f_i)$$

Example 1 Grouped Benchmark (Table 13.3)

Class Interval	f_i	x_i	$f_i x_i$
10–25	2	17.5	35.0
25–40	3	32.5	97.5
40–55	7	47.5	332.5
55–70	6	62.5	375.0
70–85	6	77.5	465.0
85–100	6	92.5	555.0
Total	**$\Sigma f_i = 30$**	—	**$\Sigma f_i x_i = 1860.0$**

$\bar{x} = 1860.0 / 30 = \mathbf{62.0}$ (Approximate mean due to midpoint grouping assumption vs raw exact mean 59.3).

⚡ Quick Box • Class Marks & Direct Mean

Q1 What is the class mark of the class interval 65–85? Can you compute it mentally in under five seconds? MENTAL CALCULATION

Q2 Why did the exact mean of raw ungrouped marks (59.3) differ slightly from the grouped direct mean (62.0) in Example 1? CORE INSIGHT

Q3 When does the Direct Method become cumbersome or tedious for a student to compute in an examination? PRACTICAL EVALUATION

The Assumed Mean Method: Centering on ’a’

The Arithmetic Shift Strategy

When class marks x_i and frequencies f_i are large, computing $f_i x_i$ is slow. We choose an **assumed mean 'a'** (usually the central class mark) and subtract it from every x_i to obtain **deviations (d_i)**:

DEVIATION DEFINITION

$$d_i = x_i - a$$

Because we subtracted 'a' from each observation, we mathematically restore it by adding 'a' back to the mean of deviations!

ASSUMED MEAN METHOD FORMULA

$$\bar{x} = a + (\Sigma f_i d_i) / (\Sigma f_i)$$

Step-by-Step Execution (Table 13.4, a = 47.5)

Class	f_i	x_i	$d_i = x_i - 47.5$	$f_i d_i$
10–25	2	17.5	–30	–60
25–40	3	32.5	–15	–45
40–55	7	47.5	0	0
55–70	6	62.5	+15	+90
70–85	6	77.5	+30	+180
85–100	6	92.5	+45	+270
Total	**30**	—	—	**$\Sigma f_i d_i = 435$**

$\bar{d} = 435 / 30 = 14.5 \Rightarrow \bar{x} = 47.5 + 14.5 =$ **62.0** (Exact same result with vastly simpler mental math!).

⚡ Quick Box • Assumed Mean Insights

Q1 If someone chooses a = 32.5 or a = 62.5 instead of 47.5 in Example 1, will the calculated final mean change? Why or why not?

ACTIVITY 1 PROOF

Q2 Why does choosing the central class mark as 'a' make calculation easiest compared to picking the very first or last class mark?

ALGEBRAIC BALANCE

Q3 What does the sum $\Sigma f_i d_i$ physically represent in terms of balance around the assumed pivot 'a'?

PHYSICAL MEANING

Pages 176–177 (Textbook)

The Step-Deviation Method: Scaling by Class Size ’h’

÷ Shrinking Deviations by Common Factor h

Notice in Table 13.4 that all deviations (−30, −15, 0, 15, 30, 45) share a common factor equal to the class size **h = 15**. Dividing by h collapses large numbers to single digits:

SCALED DEVIATION DEFINITION

$$u_i = (x_i - a) / h = d_i / h$$

Since we subtracted 'a' and divided by 'h', to retrieve $\bar{x}$ we multiply $\bar{u}$ by 'h' and add 'a'!

STEP-DEVIATION METHOD FORMULA

$$\bar{x} = a + h \times [(\Sigma f_i u_i) / (\Sigma f_i)]$$

Ultra-Compact Arithmetic (Table 13.5, h = 15)

Class	f_i	x_i	d_i	$u_i = d_i/15$	$f_i u_i$
10–25	2	17.5	−30	−2	−4
25–40	3	32.5	−15	−1	−3
40–55	7	47.5	0	0	0
55–70	6	62.5	+15	+1	+6
70–85	6	77.5	+30	+2	+12
85–100	6	92.5	+45	+3	+18
Total	**30**	—	—	—	**$\Sigma f_i u_i = 29$**

$\bar{x} = 47.5 + 15 \times (29 / 30) = 47.5 + 14.5 = \mathbf{62.0}$.

⚡ Quick Box • Step-Deviation Mastery

Q1 What single condition makes the Step-Deviation method the most convenient and fastest method to apply?

METHOD CONDITION

Q2 If class marks have intervals of width 10 and deviations are −20, −10, 0, 10, 20, what are the values of u_i?

RAPID SCALING

Q3 Does the step-deviation formula still hold if the chosen divisor 'h' is a common factor of deviations rather than the class width?

MATHEMATICAL GENERALIZATION

Method Selection & Handling Unequal Class Sizes

Example 2: Female Teachers (Table 13.7)

Percentage distribution across 35 States/UTs. Tested across all three methods with $a = 50$, $h = 10$:

- ✓ **Direct:** $\bar{x} = 1390 / 35 =$ **39.71%**
- ✓ **Assumed Mean:** $\bar{x} = 50 + (-360 / 35) = 50 - 10.29 =$ **39.71%**
- ✓ **Step-Deviation:** $\bar{x} = 50 + 10 \times (-36 / 35) =$ **39.71%**

Golden Theorem: All three methods yield the identical mathematical mean!

Example 3: Bowlers' Wickets (Unequal Sizes)

Intervals: 20–60 (h=40), 60–100 (h=40), 100–150 (h=50), 150–250 (h=100), 250–350 (h=100), 350–450 (h=100).

Class sizes vary, but deviations d_i from $a = 200$ (−160, −120, −75, 0, 100, 200) share common factors. Choosing a suitable divisor **h = 20** enables Step-Deviation:

$\bar{x} = 200 + 20 \times (-106 / 45) = 200 - 47.11 = 152.89$ wickets

Significance: An average bowler in these 45 instances claims ~153 wickets.

⚡ Quick Box • Mean Strategy & Unequal Classes

Q1 When class intervals have unequal widths, can the Step-Deviation method still be used? What rule guides the choice of 'h'?

METHOD FLEXIBILITY

Q2 If numerical values of f_i and x_i are small numbers like 1, 2, 4 and 10, 12, 14, which method should you pick and why?

EFFICIENCY CHOICE

Q3 What real-world meaning does a mean of 39.71% female teachers convey across rural schools of India?

CONTEXTUAL INTERPRETATION

Mode of Grouped Data: The Modal Class Formula

Locating the Modal Class

In grouped data, we cannot spot the single most frequent observation by sight. Instead, we locate the **modal class** — the class interval having the **maximum frequency**.

GROUPED MODE FORMULA

$$\text{Mode} = l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$$

Key Parameter Symbols Defined

- ✓ **l:** Lower limit of the modal class.
- ✓ **h:** Size of the modal class interval.
- ✓ **f_1:** Frequency of the modal class.
- ✓ **f_0:** Frequency of the class *preceding* the modal class.
- ✓ **f_2:** Frequency of the class *succeeding* the modal class.

Worked Illustration: Example 5 (Family Size)

Family Size	1–3	3–5	5–7	7–9	9–11
Number of Families	7 (f_0)	**8 (f_1)**	2 (f_2)	2	1

- ✓ Maximum frequency = 8 ⇒ **Modal Class = 3–5**
- ✓ l = 3, h = 2, f_1 = 8, f_0 = 7, f_2 = 2
- ✓ Mode = 3 + [(8 – 7) / (2×8 – 7 – 2)] × 2
- ✓ Mode = 3 + [1 / (16 – 9)] × 2 = 3 + 2/7 = **3.286**

Modal Family Size ≈ 3.286 members

⚡ Quick Box • Mode of Grouped Data

Q1 In the mode formula, what do the subscripts 0, 1, and 2 in f_0, f_1, and f_2 signify in relation to the modal class?

SUBSCRIPT MEANING

Q2 Can the computed value of the mode lie outside the boundaries of the modal class? Why or why not?

BOUNDARY CHECK

Q3 What is a distribution called if two non-adjacent classes share the exact same maximum frequency?

MULTIMODAL TERMINOLOGY

Interpreting Mode vs Mean: Which Story Does Each Tell?

Example 6: 30 Students Examination Data

Peak frequency $f_1 = 7$ is in class **40–55** (preceding $f_0 = 3$, succeeding $f_2 = 6$, $l = 40$, $h = 15$):

$$\text{Mode} = 40 + [(7 - 3) / (14 - 3 - 6)] \times 15 = 40 + 12 = 52 \text{ marks}$$

Recall from Example 1 that the arithmetic **Mean = 62 marks**.

The Crucial Real-World Contrast

Mode = 52 marks: The single most common/popular mark cluster achieved by the highest number of students.

Mean = 62 marks: The overall academic average performance score when total marks are pooled equally.

Demand of the Situation: Which to Pick?

- ✓ **Pick Mean When:** You want an overall performance benchmark representing every single individual observation (e.g., school pass average, per capita income).
- ✓ **Pick Mode When:** You need to identify the most popular, recurring, or commercially demanded item:
  - ✓ Most common shoe size sold by a retailer.
  - ✓ Most watched prime-time television broadcast.
  - ✓ Most preferred automobile color on the road.
- ✓ **Remember:** Mode can be less than, equal to, or greater than the mean depending on distribution skewness!

⚡ Quick Box • Mean vs Mode In Practice

Q1 A shoe shop owner wants to decide which shoe sizes to order in bulk from the manufacturer. Should the owner calculate the mean or the mode? BUSINESS DECISION

Q2 In Example 6, why was the mode (52) significantly lower than the mean (62)? What does that indicate about the high scorers? DATA SKEWNESS

Q3 True or False: "The mode of any given dataset is always strictly less than its arithmetic mean." Explain. FACT CHECK

Pages 188–192 (Textbook)

Cumulative Frequency: Less Than vs More Than Types

'Less Than' Type Distribution

Cumulates frequencies from the **lowest class upwards**. Uses **upper class limits** as boundary benchmarks:

Marks Benchmark	Cumulative Frequency (cf)
Less than 10	5
Less than 20	5 + 3 = 8
Less than 30	8 + 4 = 12
Less than 40	12 + 3 = 15
Less than 50	15 + 3 = 18
... up to Less than 100	Total n = **53**

Standard format used directly in the grouped median formula!

'More Than' Type Distribution

Starts with total observations n and subtracts successive class counts. Uses **lower class limits**:

Marks Benchmark	Cumulative Frequency (cf)
More than or equal to 0	53 (All students)
More than or equal to 10	53 – 5 = 48
More than or equal to 20	48 – 3 = 45
More than or equal to 30	45 – 4 = 41
More than or equal to 40	41 – 3 = 38
... More than or equal to 90	8

Essential for survival curves, cutoff screenings, and reverse ogives.

⚡ Quick Box • Cumulative Frequency Insights

Q1 Which class limits are paired with cumulative frequencies in a 'less than' distribution? Which limits are used in 'more than'?

LIMIT ASSOCIATION

Q2 What is always the very last number in the cumulative frequency column of a standard 'less than' frequency table?

TOTAL SUM CHECK

Q3 If a student has a 'less than' cumulative table, how do they recover the individual frequency of any specific class interval?

REVERSE RECOVERY

Locating the Median Class & Applying the Formula

The (n/2) Median Class Criterion

Compute **n/2** (half of total observations). Inspect the cumulative frequency (cf) column and locate the class whose cf is **just greater than (and nearest to) n/2**. This is the **Median Class**!

GROUPED MEDIAN FORMULA

$$\text{Median} = l + \left[\left(\frac{n}{2} - cf\right) / f\right] \times h$$

Crucial Distinction: 'cf' vs 'f'

- ✓ **cf:** Cumulative frequency of the class *PRECEDING* the median class.
- ✓ **f:** Simple frequency of the median class itself.
- ✓ **l:** Lower limit of the median class; **h:** Class width.

Textbook Benchmark: Table 13.15 (53 Students)

Total $n = 53 \Rightarrow n/2 = 53/2 = \mathbf{26.5}$

Looking at cf column: 5, 8, 12, 15, 18, 22, **29**... The cf just greater than 26.5 is 29, corresponding to class **60–70**!

- ✓ Median Class = 60–70 ⇒ $l = 60$, $h = 10$
- ✓ Preceding cf = 22, Frequency $f = 7$
- ✓ Median = $60 + [(26.5 - 22) / 7] \times 10 = 60 + 45/7 = \mathbf{66.4}$

Median = 66.4 marks (50% scored above, 50% below)

⚡ Quick Box • Median Class & Parameters

Q1 In the median formula, why do we use the cumulative frequency of the PRECEDING class rather than the median class itself? — FORMULA LOGIC

Q2 If total observations n = 60, what threshold value do you search for in the cumulative frequency column to locate the median class? — THRESHOLD CHECK

Q3 What does the resulting median mark of 66.4 tell us about the student cohort in plain language? — MEDIAN MEANING

Converting 'Less Than' Data & Finding Missing Frequencies

Example 7: Heights of 51 Girls

Given 'Less than 140' (4), 'Less than 145' (11), 'Less than 150' (29)... Convert to continuous class intervals first:

- ✓ Below 140: f = 4 | 140–145: f = 11 – 4 = **7**
- ✓ 145–150: f = 29 – 11 = **18** | 150–155: f = 40 – 29 = **11**
- ✓ n/2 = 51/2 = 25.5 ⇒ Median Class = **145–150**
- ✓ Median = 145 + [(25.5 – 11) / 18] × 5 = 145 + 4.03 = **149.03 cm**

Median Height = 149.03 cm

Example 8: Solving for x and y (n = 100)

Given total frequency = 100 and known **Median = 525**:

- ✓ Total sum: 76 + x + y = 100 ⇒ **x + y = 24** (Eq 1)
- ✓ Median 525 lies in class **500–600** ⇒ l = 500, f = 20, h = 100, preceding cf = 36 + x
- ✓ 525 = 500 + [(50 – (36 + x)) / 20] × 100
- ✓ 25 = (14 – x) × 5 ⇒ 14 – x = 5 ⇒ **x = 9**
- ✓ From Eq 1: 9 + y = 24 ⇒ **y = 15**

Solution: x = 9, y = 15

⚡ Quick Box • Cumulative Table Conversions

Q1 If a problem states "Median = 525", how do you instantly identify the median class without looking at n/2?

DIRECT VALUE MATCH

Q2 When converting a 'less than' cumulative table into class intervals, how do you find the frequency of the interval 145–150?

SUBTRACTIVE DIFFERENCE

Q3 Why does a missing frequency problem always require two distinct equations to find both unknown variables x and y?

ALGEBRAIC SYSTEM

Converting Inclusive (Discontinuous) Classes

Mandatory NCERT Reader's Note

The formulas for mode and median assume that class boundaries are **strictly continuous**! If classes have gaps (e.g. 118–126, 127–135), you **MUST** convert them before calculating.

BOUNDARY ADJUSTMENT RULE

Gap = 127 − 126 = 1 ⇒ Correction Factor = 1/2 = 0.5

Subtract 0.5 from Lower Limit & Add 0.5 to Upper Limit

Notice how 118–126 becomes **117.5–126.5**, and 127–135 becomes **126.5–135.5**. The boundary point 126.5 is now seamlessly shared!

Exercise 13.3 Q4: Plant Leaves Benchmark

Original (Gapped)	Continuous Interval	Leaves (f)	cf
118–126	117.5–126.5	3	3
127–135	126.5–135.5	5	8
136–144	135.5–144.5	9	17
145–153	**144.5–153.5**	**12**	**29**
154–162	153.5–162.5	5	34
163–171	162.5–171.5	4	38
172–180	171.5–180.5	2	40

Total n = 40 ⇒ n/2 = 20 ⇒ Median Class = **144.5–153.5** (l = 144.5, not 145!).

⚡ Quick Box • Class Continuity Check

Q1 What major calculation error occurs in finding the median if a student forgets to convert 118–126 into 117.5–126.5?

LOWER LIMIT ERROR

Q2 If classes are 1–5, 6–10, 11–15, what is the gap size, what is the correction factor, and what are the continuous classes?

RAPID CONVERSION

Q3 Does the mean of grouped data strictly require continuous class limits in the same mandatory way as median and mode?

COMPARATIVE RULE

Comparing Measures & The Empirical Formula

THE GOLDEN EMPIRICAL RELATIONSHIP

3 Median = Mode + 2 Mean

Alternative Form: Mode = 3 Median – 2 Mean

Instant 2-Second Exam Solver

Whenever any two measures of central tendency are known for a moderately skewed distribution, the third measure can be computed instantly without re-analyzing the raw data table!

When to Use Which Measure? (NCERT Guide)

- ✓ **Mean:** Best when all observations matter equally and the data has no extreme outliers. Perfect for comparing overall school or factory performances.
- ✓ **Median:** Best when extreme outliers distort the average. Identifies a *typical* observation (e.g. median household wage, typical worker productivity).
- ✓ **Mode:** Best for identifying the *most popular or most demanded* item in business, manufacturing, and commerce.

⚡ Quick Box • Empirical Relationship Drills

Q1 If a distribution has Mean = 20 and Median = 22, calculate its Mode in 5 seconds using the empirical relationship.

MENTAL CALCULATION

Q2 Why does the presence of extreme outliers (like a billionaire living in a small village) make Median a better measure of central tendency than Mean?

OUTLIER IMPACT

Q3 In a perfectly symmetrical distribution, what is the exact mathematical relationship between Mean, Median, and Mode?

SYMMETRY PROPERTY

Exercise 13.1 Q3: Missing Frequency 'f' (Mean Allowance)

EXERCISE 13.1 • QUESTION 3

Daily Pocket Allowance of Children ($\bar{x}$ = ₹18)

Given: Mean pocket allowance = ₹18. Find missing frequency f.

Allowance (₹)	f_i	x_i	$d_i = x_i - 18$	$f_i d_i$
11–13	7	12	−6	−42
13–15	6	14	−4	−24
15–17	9	16	−2	−18
17–19	13	18 (a)	0	0
19–21	f	20	+2	+2f
21–23	5	22	+4	+20
23–25	4	24	+6	+24

Solution Steps via Assumed Mean a = 18

- ✓ Total frequency: $\Sigma f_i = 7 + 6 + 9 + 13 + f + 5 + 4 = \mathbf{44 + f}$
- ✓ Sum of $f_i d_i$: $(-42 - 24 - 18) + (2f + 20 + 24)$
  $= -84 + 44 + 2f = \mathbf{2f - 40}$
- ✓ Assumed Mean Formula: $\bar{x} = a + (\Sigma f_i d_i) / (\Sigma f_i)$
- ✓ $18 = 18 + [(2f - 40) / (44 + f)]$
- ✓ $0 = (2f - 40) / (44 + f) \Rightarrow 2f - 40 = 0 \Rightarrow \mathbf{2f = 40 \Rightarrow f = 20}$

Missing Frequency f = 20 children

Page 186 (Textbook)

Exercise 13.2 Q2: Modal Lifetimes of Electrical Components

EXERCISE 13.2 • QUESTION 2

Lifetimes of 225 Electrical Components

Find the modal lifetime of the observed components from the distribution:

Lifetimes (hours)	Frequency
0–20	10
20–40	35
40–60	52 (f_0)
60–80	**61 (f_1) — Peak**
80–100	38 (f_2)
100–120	29

Modal Computation Steps

- ✓ Maximum frequency = 61 ⇒ **Modal Class = 60–80**
- ✓ Lower limit l = 60, class size h = 20
- ✓ $f_1 = 61$, preceding $f_0 = 52$, succeeding $f_2 = 38$
- ✓ Mode = $l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$
- ✓ Mode = $60 + [(61 - 52) / (2\times61 - 52 - 38)] \times 20$
- ✓ Mode = $60 + [9 / (122 - 90)] \times 20 = 60 + [9 / 32] \times 20$
- ✓ Mode = $60 + 45/8 = 60 + 5.625 =$ **65.625 hours**

Modal Lifetime = 65.625 hours

Exercise 13.3 Q2: Double Missing Frequencies (x & y)

EXERCISE 13.3 • QUESTION 2

Given Median = 28.5 & Total Frequency = 60

Class	Frequency	Cumulative Frequency (cf)
0–10	5	5
10–20	x	5 + x
20–30	**20**	**25 + x**
30–40	15	40 + x
40–50	y	40 + x + y
50–60	5	45 + x + y
Total	**60**	—

System of Equations Solution

- ✓ Total sum: 45 + x + y = 60 ⇒ **x + y = 15** (Eq 1)
- ✓ Median = 28.5 lies in class **20–30**!
- ✓ l = 20, h = 10, f = 20, preceding cf = 5 + x, n/2 = 30
- ✓ Median = l + [((n/2) − cf) / f] × h
- ✓ 28.5 = 20 + [(30 − (5 + x)) / 20] × 10
- ✓ 8.5 = (25 − x) / 2 ⇒ 25 − x = 17 ⇒ **x = 8**
- ✓ From Eq 1: 8 + y = 15 ⇒ **y = 7**

Values: x = 8 and y = 7

Pages 200–201 (Textbook)

Chapter 13 Master Formula & Concept Checklist

The Three Mean Methods ($\bar{x}$)

- ✓ **Direct Method:**
  $\bar{x} = (\Sigma f_i x_i) / (\Sigma f_i)$
- ✓ **Assumed Mean Method:**
  $\bar{x} = a + (\Sigma f_i d_i) / (\Sigma f_i)$ [where $d_i = x_i - a$]
- ✓ **Step-Deviation Method:**
  $\bar{x} = a + h \times [(\Sigma f_i u_i) / (\Sigma f_i)]$ [where $u_i = (x_i - a)/h$]

Mode, Median & Empirical Law

- ✓ **Mode of Grouped Data:**
  Mode = $l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$
- ✓ **Median of Grouped Data:**
  Median = $l + [((n/2) - cf) / f] \times h$
- ✓ **Empirical Relationship:**
  3 Median = Mode + 2 Mean

NCERT CLASS X MATHEMATICS

Thank You!

"Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write. Turn data into clarity and wisdom."

Akash Srivastva (www.akashmaths.online)

Engineered for high-impact classroom presentation, conceptual mastery, and board examination excellence.

Akash Srivastva (www.akashmaths.online)